THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of composite numbers from 1 to 50 (both included) is: 1

(a) 33 **(b)** 36 **(c)** 35 **(d)** 34

2. The smallest number divisible by each of 10, 15 and 20 is: 1

(a) 61 **(b)** 60 **(c)** 70 **(d)** 59

3. The greatest number that divides 60, 90 and 150 leaving no remainder is: 1

(a) 30 **(b)** 32 **(c)** 29 **(d)** 31

4. For the 3-digit number 2d5, the digit d that makes it divisible by 9 (smallest such d) is: 1

(a) 2 **(b)** 11 **(c)** 4 **(d)** 3

5. Which is the greatest 3-digit number that leaves remainder 4 when divided by 9? 1

(a) 985 **(b)** 1003 **(c)** 994 **(d)** 995

6. The sum of the first 4 prime numbers after 10 is: 1

(a) 61 **(b)** 62 **(c)** 59 **(d)** 60

7. The largest prime number less than 150 is: 1

(a) 150 **(b)** 149 **(c)** 148 **(d)** 151

8. The product of the two smallest prime factors of 385 is: 1

(a) 12 **(b)** 35 **(c)** 77 **(d)** 55

9. **Assertion (A):** 91 is a composite number. 1

Reason (R): Every number that is not divisible by 2, 3 or 5 is a prime number.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** 51 is a prime number. 1

Reason (R): A prime number has exactly two factors.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Show that the product of 4 consecutive numbers starting from 23 is divisible by 4. 2

12. Find the prime factors of 66 and the smallest number to multiply to make it divisible by 40. 2

13. Find the smallest number divisible by 9, 15 and 21. 2

14. Show that the product of 3 consecutive numbers starting from 31 is divisible by 3. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Three traffic lights change every 36, 60 and 90 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

16. Find the greatest number that divides 95 and 119 leaving remainder 7 in each case. 3

17. Find the smallest number which when divided by 8, 9 and 12 leaves remainder 4 each time. 3

OR

Three traffic lights change every 40, 50 and 100 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

18. Find the greatest number that divides 200 and 242 leaving remainder 2 in each case. 3

OR

Find the smallest number which when divided by 12, 15 and 20 leaves remainder 7 each time. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A hall floor is 120 cm by 84 cm and is to be covered with square tiles of the largest possible size, with no tile cut. 5

(i) Write 120 and 84 as products of prime factors.

(ii) Find the side of the largest square tile.

(iii) How many tiles fit along the length and along the breadth?

(iv) Find the total number of tiles and check it by using areas.

20. A rectangular garden is 90 m by 54 m. Square flower beds of the largest possible equal size are to be made covering the garden. Find 5

(i) the side of each bed,

(ii) the number of beds,

(iii) the cost of fencing all beds at ₹3 per metre if each bed is fenced separately.

OR

Find the smallest 4-digit number divisible by 14, 16 and 24. Also find the largest 4-digit number divisible by all three. 5

END OF QUESTION PAPER — Total: 40 Marks